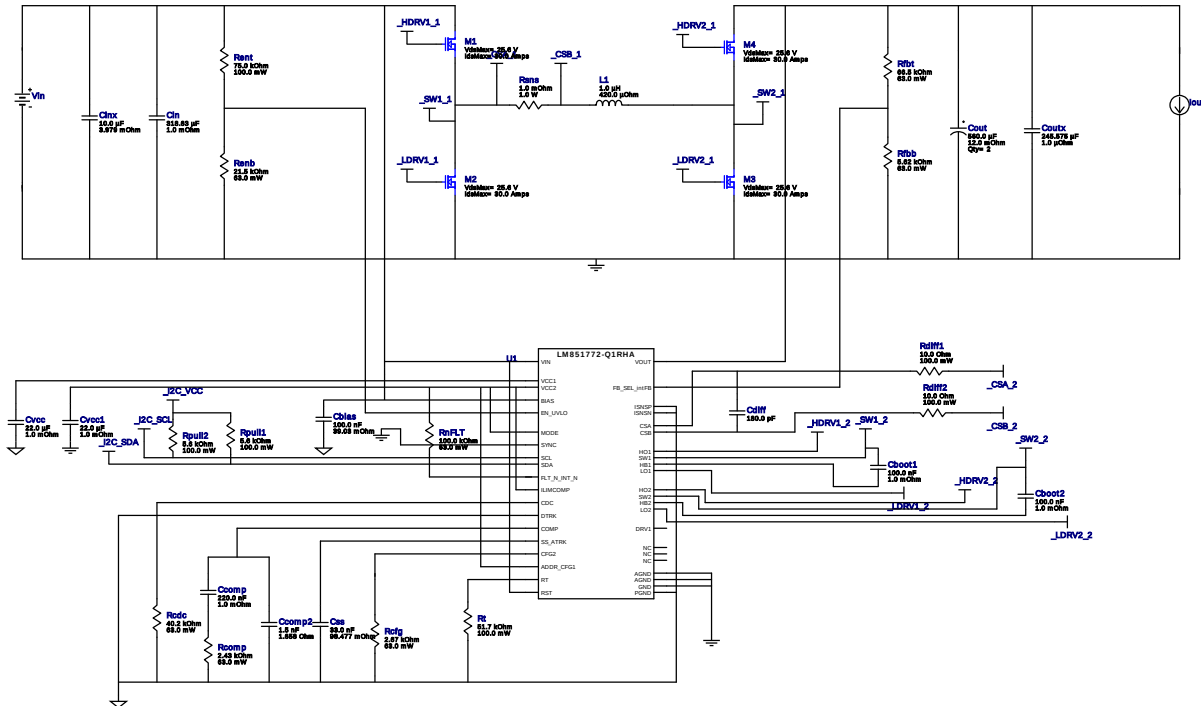


WEBENCH® Design Report

Design : 5 LM851772QRHARQ1
LM851772QRHARQ1 6V-42V to 12.80V @ 20A



1. The LM851772-Q1 features AutoMode for better light load efficiency. The operating point calculated in AutoMode configuration are estimates

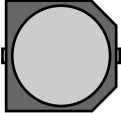
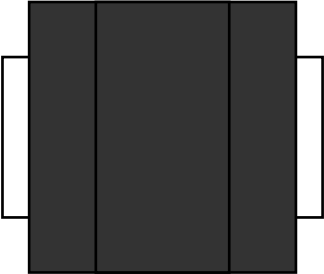
Design Alerts

Component Selection Information

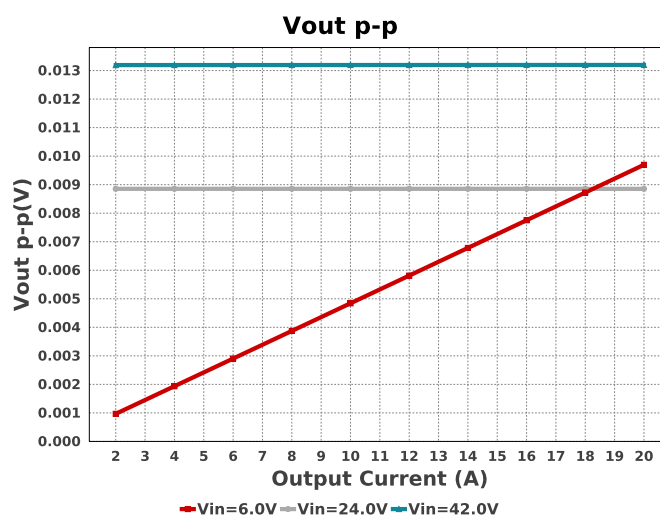
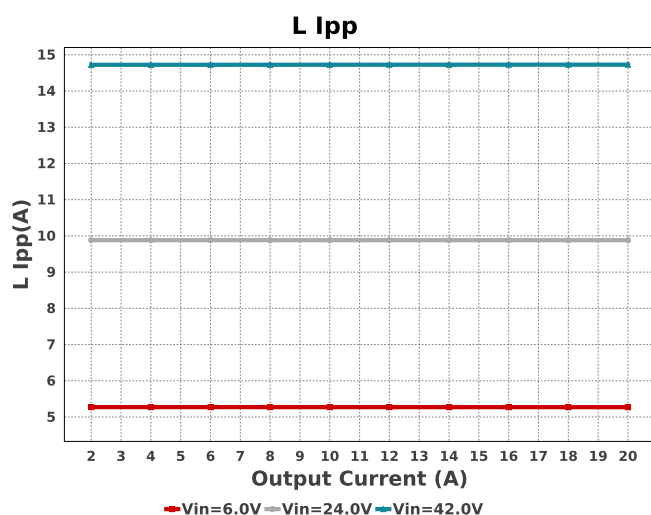
The LM851772-Q1 is qualified for Automotive applications. All passives and other components selected in this design may not be qualified for Automotive applications. The user is required to verify that all components in the design meet the qualification and safety requirements for their specific application.

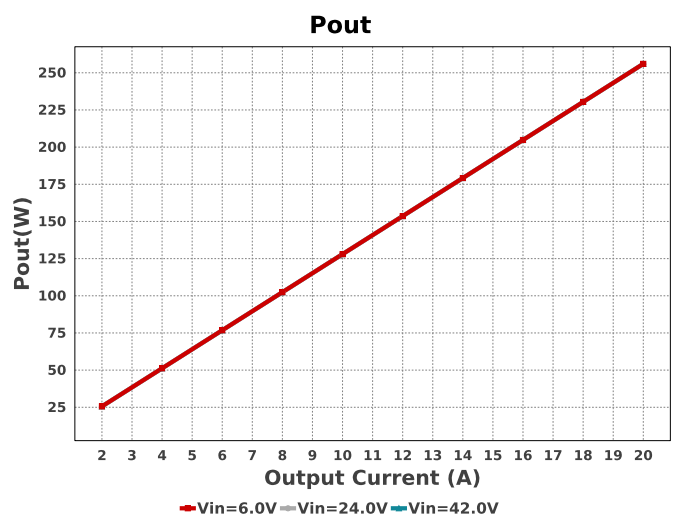
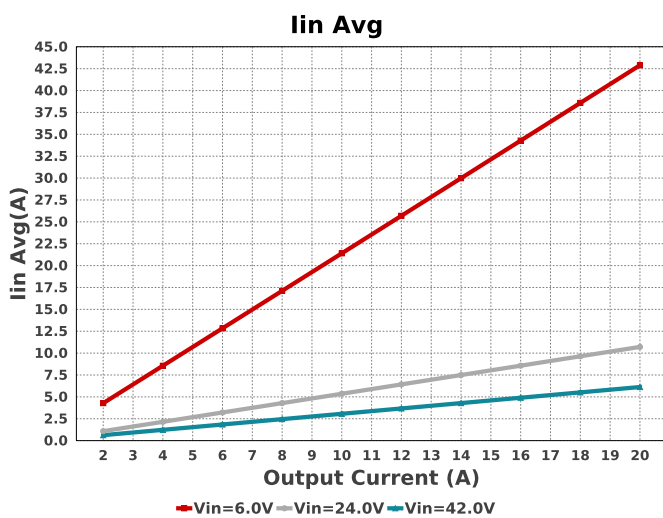
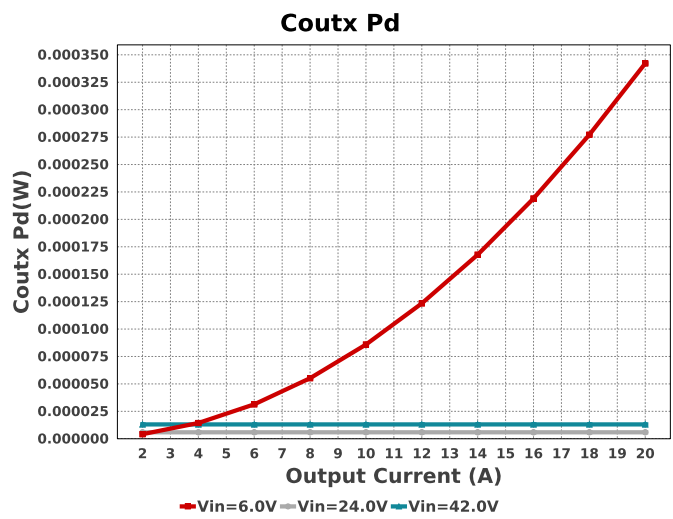
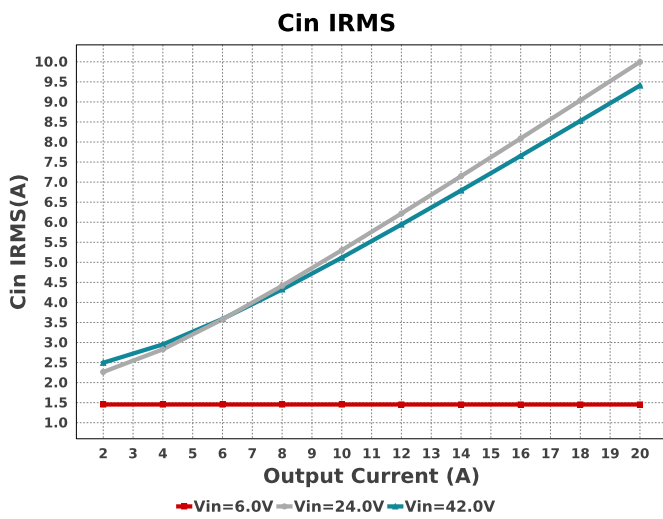
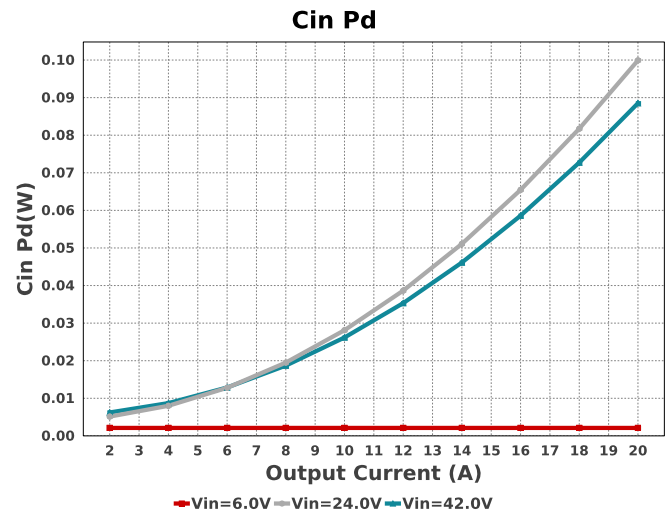
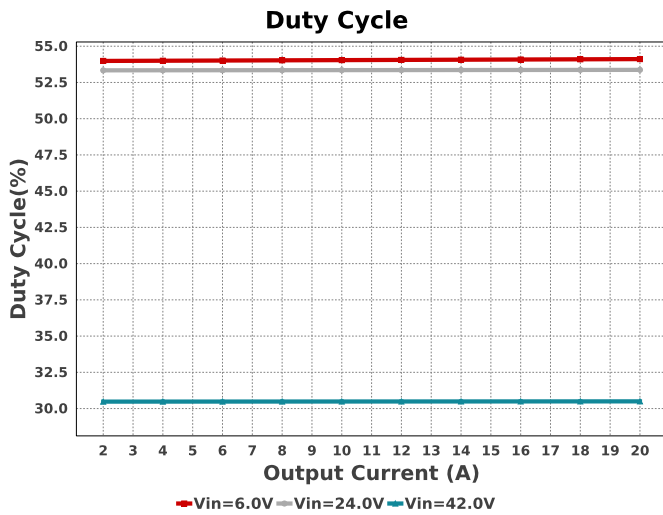
Electrical BOM

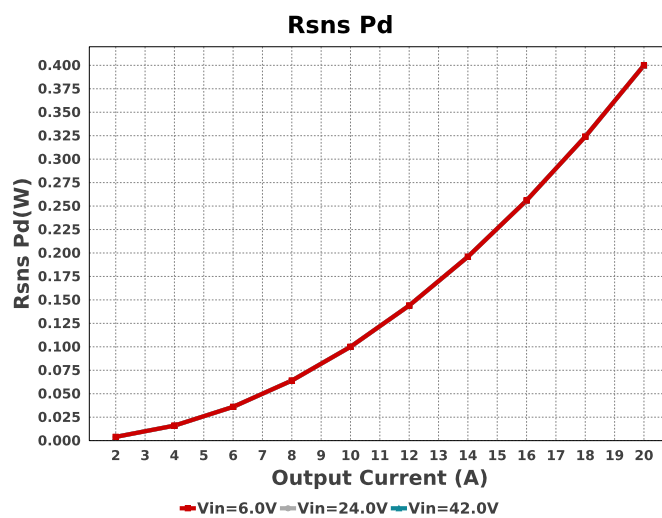
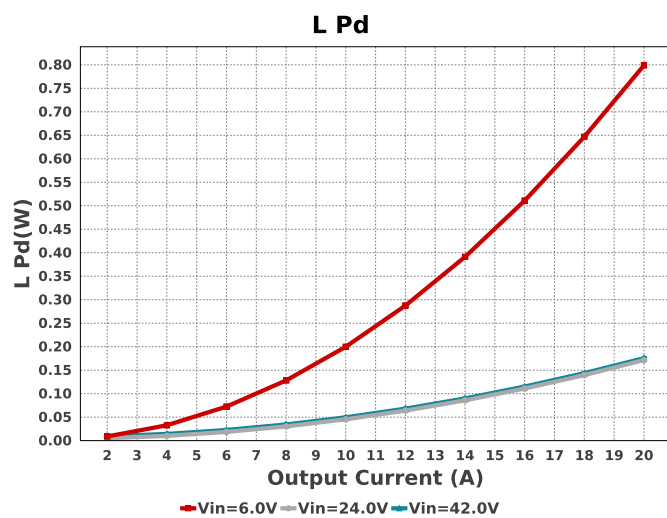
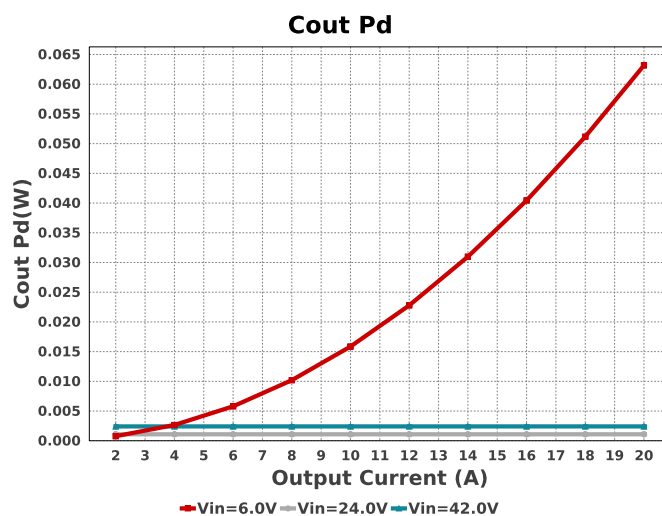
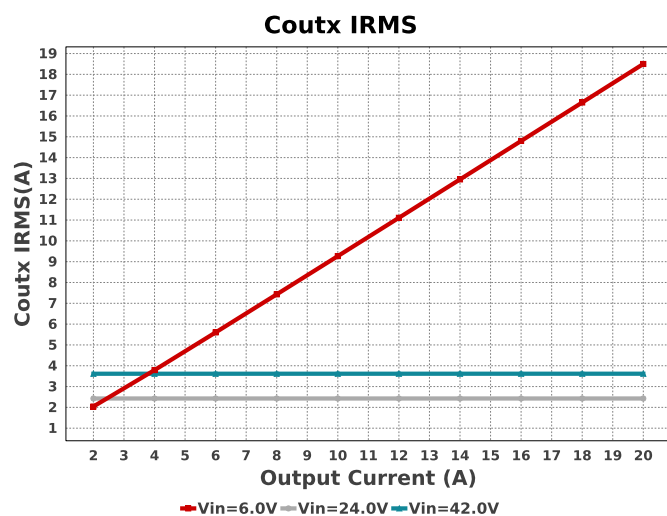
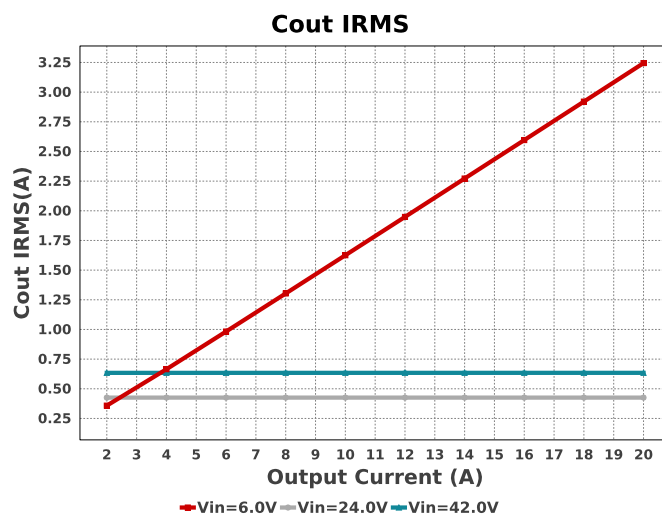
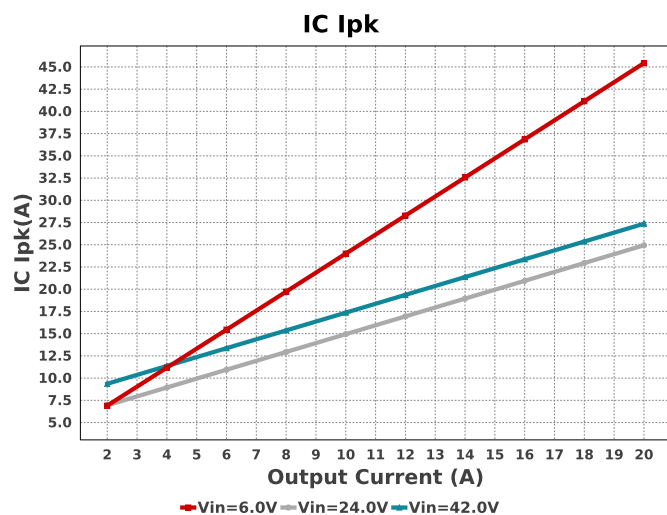
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cbias	TDK	C1005X7R1H104K050BB Series= X7R	Cap= 100.0 nF ESR= 39.03 mOhm VDC= 50.0 V IRMS= 0.0 A	1	\$0.02	0402 3 mm ²
Cboot1	Taiyo Yuden	EMK107B7104KA-T Series= X7R	Cap= 100.0 nF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	\$0.01	0603 5 mm ²
Cboot2	Taiyo Yuden	EMK107B7104KA-T Series= X7R	Cap= 100.0 nF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	\$0.01	0603 5 mm ²
Ccomp	MuRata	GRM155R71C224KA12D Series= X7R	Cap= 220.0 nF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	\$0.02	0402 3 mm ²
Ccomp2	TDK	CGA2B2X7R1H152K050BA Series= X7R	Cap= 1.5 nF ESR= 1.6581 Ohm VDC= 50.0 V IRMS= 254.831 mA	1	\$0.01	0402 3 mm ²

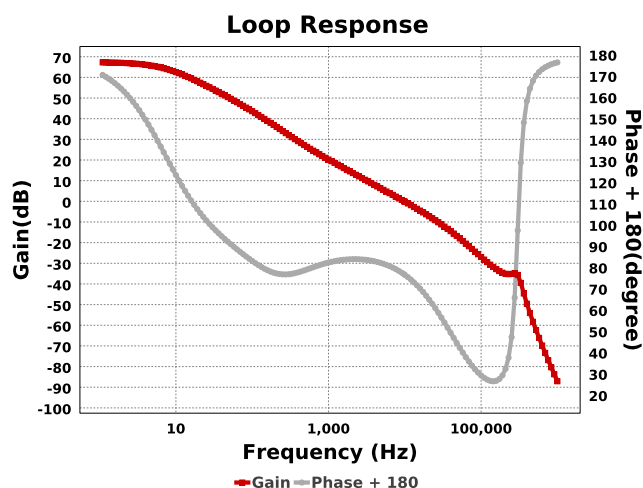
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Cdiff	Kemet	C0805C181K5GACTU Series= C0G/NP0	Cap= 180.0 pF VDC= 50.0 V IRMS= 0.0 A	1	\$0.01	 0805 7 mm ²
Cin	CUSTOM	CUSTOM Series= ?	Cap= 318.63 uF ESR= 1.0 mOhm VDC= 50.4 V IRMS= 4.8843 A	1	NA	CUSTOM 0 mm ²
Cinx	Taiyo Yuden	MSASU32MSB5106KPNA01 Series= X5R	Cap= 10.0 uF ESR= 3.979 mOhm VDC= 50.0 V IRMS= 3.4821 A	1	\$0.16	 1210 15 mm ²
Cout	Panasonic	20SVPF560M Series= SVPF	Cap= 560.0 uF ESR= 12.0 mOhm VDC= 20.0 V IRMS= 5.4 A	2	\$1.33	 CAPSMT_62_F12 151 mm ²
Coutx	CUSTOM	CUSTOM Series= ?	Cap= 245.575 uF ESR= 1.0 uOhm VDC= 25.6 V IRMS= 1.06155 A	1	NA	CUSTOM 0 mm ²
Css	TDK	CGA2B3X7R1H333K050BB Series= X7R	Cap= 33.0 nF ESR= 98.477 mOhm VDC= 50.0 V IRMS= 530.78 mA	1	\$0.01	 0402 3 mm ²
Cvcc	MuRata	GRM188R60J226MEA0D Series= X5R	Cap= 22.0 uF ESR= 1.0 mOhm VDC= 6.3 V IRMS= 6.0 A	1	\$0.04	 0603 5 mm ²
Cvcc1	MuRata	GRM188R60J226MEA0D Series= X5R	Cap= 22.0 uF ESR= 1.0 mOhm VDC= 6.3 V IRMS= 6.0 A	1	\$0.04	 0603 5 mm ²
L1	Coiltronics	HC3-1R0-R	L= 1.0 µH 420.0 µOhm	1	\$6.88	 HC3 874 mm ²
M1	NA	IdealFET	VdsMax= 25.6 V IdsMax= 30.0 Amps	1	NA	KCS0003B 80 mm ²
M2	NA	IdealFET	VdsMax= 25.6 V IdsMax= 30.0 Amps	1	NA	KCS0003B 80 mm ²
M3	NA	IdealFET	VdsMax= 25.6 V IdsMax= 30.0 Amps	1	NA	KCS0003B 80 mm ²
M4	NA	IdealFET	VdsMax= 25.6 V IdsMax= 30.0 Amps	1	NA	KCS0003B 80 mm ²
Rcdc	Vishay-Dale	CRCW040240K2FKED Series= CRCW..e3	Res= 40.2 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	 0402 3 mm ²
Rcfg	Vishay-Dale	CRCW04022K67FKED Series= CRCW..e3	Res= 2.67 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	 0402 3 mm ²
Rcfg	Vishay-Dale	CRCW04022K67FKED Series= CRCW..e3	Res= 2.67 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	 0402 3 mm ²

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Rcomp	Vishay-Dale	CRCW04022K43FKED Series= CRCW..e3	Res= 2.43 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
Rdiff1	Vishay-Dale	CRCW060310R0FKEA Series= CRCW..e3	Res= 10.0 Ohm Power= 100.0 mW Tolerance= 1.0%	1	\$0.01	0603 5 mm ²
Rdiff2	Vishay-Dale	CRCW060310R0FKEA Series= CRCW..e3	Res= 10.0 Ohm Power= 100.0 mW Tolerance= 1.0%	1	\$0.01	0603 5 mm ²
Renb	Vishay-Dale	CRCW040221K5FKED Series= CRCW..e3	Res= 21.5 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
Rent	Vishay-Dale	CRCW060375K0FKEA Series= CRCW..e3	Res= 75.0 kOhm Power= 100.0 mW Tolerance= 1.0%	1	\$0.01	0603 5 mm ²
Rfbb	Vishay-Dale	CRCW04025K62FKED Series= CRCW..e3	Res= 5.62 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
Rfbb1	Vishay-Dale	CRCW04025K62FKED Series= CRCW..e3	Res= 5.62 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
Rfbt	Vishay-Dale	CRCW040266K5FKED Series= CRCW..e3	Res= 66.5 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
Rfbt1	Vishay-Dale	CRCW040210K7FKED Series= CRCW..e3	Res= 10.7 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
RnFLT	Vishay-Dale	CRCW0402100KFKED Series= CRCW..e3	Res= 100.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
Rpull1	Yageo	RC0603FR-075K6L Series= ?	Res= 5.6 kOhm Power= 100.0 mW Tolerance= 1.0%	1	\$0.01	0603 5 mm ²
Rpull2	Yageo	RC0603FR-075K6L Series= ?	Res= 5.6 kOhm Power= 100.0 mW Tolerance= 1.0%	1	\$0.01	0603 5 mm ²
Rsns	Vishay-Dale	WSL20101L000FEA18 Series= WSL...18	Res= 1.0 mOhm Power= 1.0 W Tolerance= 1.0%	1	\$0.73	2010 32 mm ²
Rt	Yageo	RT0603DRE0751K7L Series= RT0603	Res= 51.7 kOhm Power= 100.0 mW Tolerance= 0.5%	1	\$0.01	0603 5 mm ²
U1	Texas Instruments	LM851772QRHARQ1	Switcher	1	\$2.00	RHA0040P 64 mm ²









Operating Values

#	Name	Value	Category	Description
1.	Cin IRMS	1.452 A	Capacitor	Input capacitor RMS ripple current
2.	Cin Pd	2.108 mW	Capacitor	Input capacitor power dissipation
3.	Cout IRMS	3.325 A	Capacitor	Output capacitor RMS ripple current
4.	Cout Pd	66.349 mW	Capacitor	Output capacitor power dissipation
5.	Coutx IRMS	18.623 A	Capacitor	Output capacitor_x RMS ripple current
6.	Coutx Pd	346.82 μ W	Capacitor	Output capacitor_x power loss
7.	IC Ipk	45.861 A	IC	Peak switch current in IC
8.	IC Tolerance	10.0 mV	IC	IC Feedback Tolerance
9.	Iin Avg	42.898 A	IC	Average input current
10.	L Ipp	5.264 A	Inductor	Peak-to-peak inductor ripple current
11.	L Pd	815.35 mW	Inductor	Inductor power dissipation
12.	Cin Pd	2.108 mW	Power	Input capacitor power dissipation
13.	Cout Pd	66.349 mW	Power	Output capacitor power dissipation
14.	Coutx Pd	346.82 μ W	Power	Output capacitor_x power loss
15.	L Pd	815.35 mW	Power	Inductor power dissipation
16.	Rsns Pd	400.0 mW	Power	Current Limit Sense Resistor Power Dissipation
17.	Rsns Pd	400.0 mW	Resistor	Current Limit Sense Resistor Power Dissipation
18.	BOM Count	37	System	Total Design BOM count
19.	Cross Freq	2.133 kHz	System	Bode plot crossover frequency
20.	Duty Cycle	54.581 %	System	Duty cycle
21.	FootPrint	1.775 k mm ²	System	Total Foot Print Area of BOM components
22.	Frequency	604.449 kHz	System	Switching frequency
23.	Gain Marg	-20.192 dB	System	Bode Plot Gain Margin
24.	Iout	20.0 A	System	Iout operating point
25.	Low Freq Gain	53.677 dB	System	Gain at 1Hz
26.	Mode	CCM	System	Conduction Mode
27.	Phase Marg	78.77 deg	System	Bode Plot Phase Margin
28.	Pout	256.0 W	System	Total output power
29.	Total BOM	NA	System	Total BOM Cost
30.	Vin	6.0 V	System	Vin operating point
31.	Vout	12.8 V	System	Operational Output Voltage
32.	Vout Actual	12.833 V	System	Vout Actual calculated based on selected voltage divider resistors
33.	Vout Tolerance	2.881 %	System	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
34.	Vout p-p	9.93 mV	System	Peak-to-peak output ripple voltage

Design Inputs

Name	Value	Description
Iout	20.0	Maximum Output Current
VinMax	42.0	Maximum input voltage
VinMin	6.0	Minimum input voltage
Vout	12.8	Output Voltage
base_pn	LM851772-Q1	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature

WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of C_{in} and C_{out} , and the inductance and DC resistance of $L1$ before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

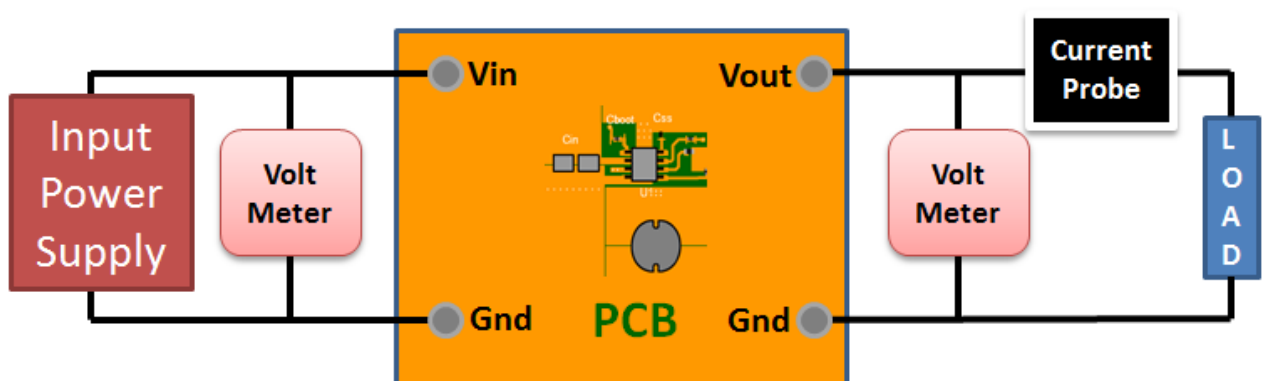
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab down to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 6.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to V_{in} and GND. Connect a digital volt meter and a load if needed to set the minimum load of the design from V_{out} and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between V_{in} and GND, a load is connected between V_{out} and GND and a current meter is connected in series between V_{out} and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

1. Master key : 8798C1FA2C1E78B872875FCFEF5BDD77[v1]
2. **LM851772-Q1** Product Folder : <https://www.ti.com/product/LM851772%2DQ1> : contains the data sheet and other resources.

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